Problem Set 3: Income Fluctuations and Ergodic Distributions

Solutions + Comments

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Roadmap

Problem 1: Two-State Income

- Discrete (employed/unemployed) income
- Three solution methods: Grid VFI, Coefficient VFI, Howard
- Ergodic distribution via transition matrix eigenvector

Problem 2: AR(1) Income

- Continuous income, Gaussian quadrature expectations
- Tensor-product spline collocation on Greville nodes
- Decision rules, Euler diagnostics, ergodic distribution

Common thread

Both problems illustrate the gap between **solve grid** (where the Bellman is enforced) and **output grid** (where distributions and moments are computed).

Problem 1: Economic Environment

Household problem with two income states $e \in \{\text{employed}, \text{unemployed}\}$:

$$E(a) = \max_{a' \geq 0} \log\big((1+r)a + 1 - a'\big) + \beta\big[(1-\sigma)E(a') + \sigma U(a')\big]$$

$$U(a) = \max_{a' \geq 0} \log\big((1+r)a + b - a'\big) + \beta\big[\lambda E(a') + (1-\lambda)U(a')\big]$$

Income and transitions

- Employed: $y = 1$, Unemployed: $y = b = 0.25$
- $\sigma = 0.10$ (job-loss rate)
- $\lambda = 0.25$ (job-finding rate)
- Fraction employed: $\lambda/(\sigma+\lambda) = 0.714$

Calibration and grid

- $\beta = 0.95$, $r = 0.04$, $a \in [0, 10]$
- Borrowing constraint: $a' \geq 0$
- 51 cubic spline breakpoints; 53 Greville collocation nodes
- Dense evaluation grid: 551 pts (quadratic spacing)

Problem 1: Solution Methods and Convergence

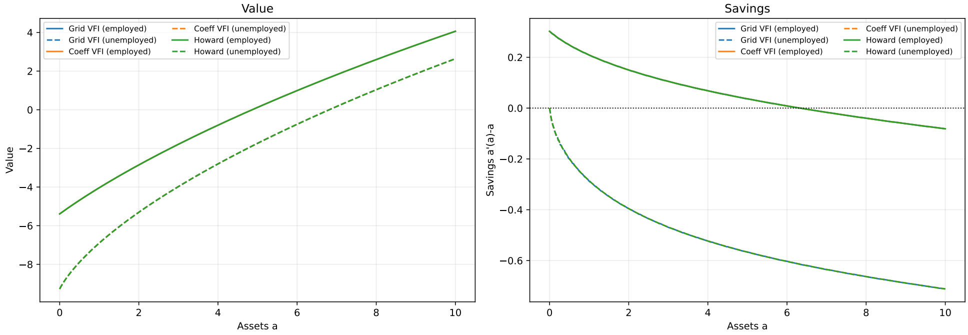
Three methods solving the same fixed point:

- **Grid VFI**: maximize Bellman at each of 551 grid points via golden-section search; interpolate continuation value with cubic spline.
- **Coefficient VFI**: represent $E(\cdot)$, $U(\cdot)$ as spline coefficients θ_e, θ_u; update $\theta^{t+1} = \Phi^{-1} v^t$ at each iteration.
- **Howard Improvement**: fix policy $a'(a)$; solve the **linear system** for coefficients directly; repeat until policy stabilizes.

Method	Iterations	Final error	Max Δ value vs Grid
Grid VFI	336	`9.70e-09`	— (benchmark)
Coefficient VFI	420	`9.50e-10`	`2.19e-04`
Howard	**7**	`2.76e-14`	`2.19e-04`

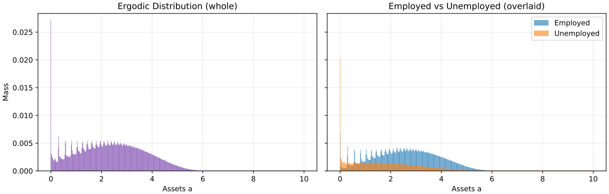
Max absolute difference in savings policy (employed): `1.22e-03` for both Coefficient VFI and Howard.

Problem 1: Value Functions and Savings Policies



Left: Value functions $E(a)$ and $U(a)$ — all three methods overlap perfectly (max diff $\sim 2 \times 10^{-4}$). **Right**: Net savings $a'(a) - a$ — employed agents save more at every wealth level; the zero-crossing identifies the **target wealth**. Both states show a region of dis-saving near the borrowing constraint.

Problem 1: Ergodic Distribution



Method: Transition matrix Q from spline basis; stationary eigenvector (unit eigenvalue).

Ergodic moments

	Value
Fraction employed	0.714
Mean assets (E)	2.60
Mean assets (U)	1.75
Mean consump. (E)	0.97
Mean consump. (U)	0.65
Stationarity error	`1.3e-17`

Takeaway: Unemployed agents hold *less* wealth on average — precautionary savings collapse near the borrowing constraint.

Problem 2: Continuous AR(1) Income — Environment

Household Bellman equation:

$$\boxed{V(a,z) = \max_{a' \geq 0} \frac{c^{1-\alpha}}{1-\alpha} + \beta \int V(a', \rho z + \sigma \varepsilon')\, dF(\varepsilon') \quad \text{s.t.} \quad c + a' = e^z + (1+r)a}$$

Income process

- AR(1) in logs: $z_{t+1} = \rho z_t + \sigma \varepsilon_{t+1}$, $\varepsilon \sim \mathcal{N}(0,1)$
- $\rho = 0.90$, $\sigma = 0.20$
- z grid: $\pm 4\, \sigma_z^\infty$ from mean, $\sigma_z^\infty = \sigma / \sqrt{1-\rho^2}$
- Expectations: Gauss–Hermite quadrature ($k = 7$ nodes)

Calibration

- $\beta = 0.95$, $r = 0.02$, $\alpha = 1.5$ (CRRA)
- $a \in [0, 50]$, 51 breakpoints (u^2 spacing)
- 15 z breakpoints, total **901 collocation nodes**
- Greville abscissae as solve grid; dense uniform grid for output

Problem 2: Grid Design — Solve Grid vs. Output Grid

Asset grid (a-axis): 51 breakpoints via u^2 clustering $\Rightarrow$ 53 Greville nodes. Dense coverage at low a.

Income grid (z-axis): 15 breakpoints $\Rightarrow$ 17 Greville nodes. Range: $\pm 4\,\sigma_z^\infty$.

Total: $53 \times 17 = 901$ collocation states where the Bellman is solved.

Key distinction

Solve grid: Greville nodes (901 pts). Bellman enforced *only* here; spacing set by the spline basis.

Output grid: dense uniform grid (e.g. 101×31) for distribution, moments, and Euler diagnostics. Policies *evaluated* here via spline.

Mixing the two is the most common source of distribution errors.

Problem 2: Solution Methods and Convergence

Coefficient VFI: solve $\theta^{t+1} = \Phi^{-1} v(a, z; \theta^t)$ at 901 collocation nodes.

Howard: fix optimal policy; update θ by solving the linear system

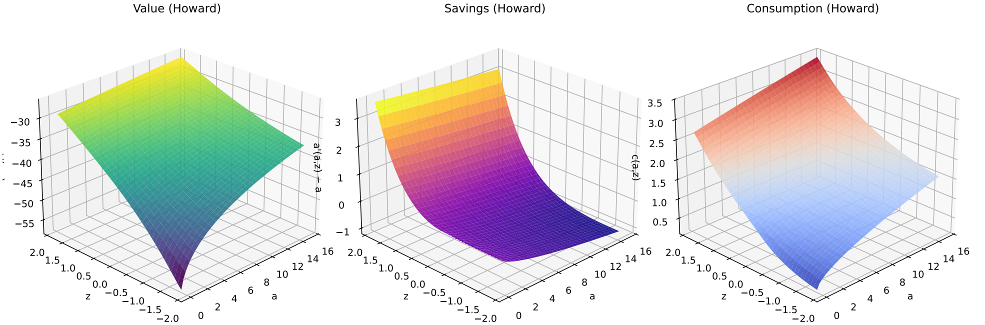
$$\left[\Phi - \beta \sum_{k=1}^{K} w_k \Phi(a'(\cdot), \rho z + \sigma \varepsilon_k)\right] \theta = u(c)$$

until coefficients converge; then recompute policy.

Method	**Iter**	**Error**	**Time**	**ΔV vs Grid**
Grid VFI (81×25)	285	`9.56e-07`	2.6 s	benchmark
Coeff VFI (901 nodes)	285	`9.56e-07`	45 s	<`1e-04`
Howard (901 nodes)	**6**	`2.27e-08`	fast	<`1e-04`

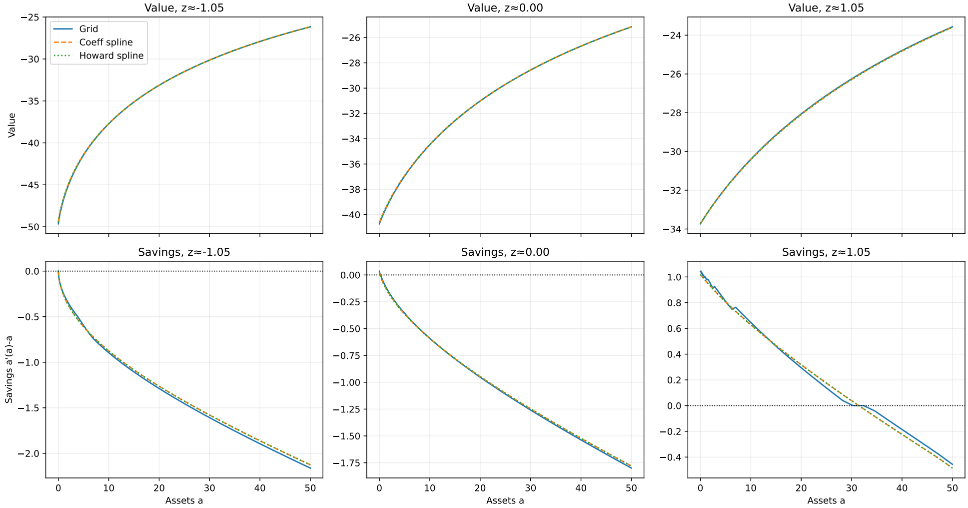
Takeaway: Howard: 6 iterations vs. 285 (VFI). Same fixed point; far fewer maximization calls.

Problem 2: Policy Surfaces (Howard Solution)



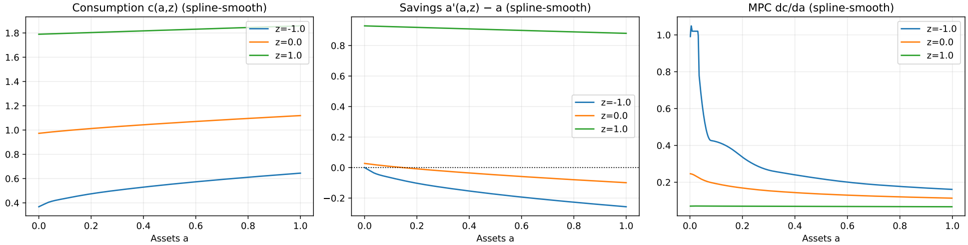
Left: $V(a, z)$ concave in a, increasing and convex in z (log-income curvature). **Middle**: Net savings $a'(a, z) - a$ increasing in both a and z; negative for low-a, low-z agents (dis-savers at the constraint). **Right**: Consumption $c(a, z)$ increasing and concave in a; agents with high z consume a larger level but also a smaller fraction of resources.

Problem 2: Method Comparison at Selected z Values



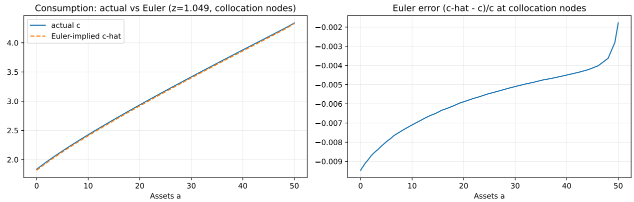
Top row: value functions over full asset range $a \in [0, 50]$; **Bottom row**: net savings $a'(a, z) - a$. Columns: $z \in \{-1, 0, 1\}$ (low, median, high income). All three methods (Grid

Problem 2: Decision Rules and Marginal Propensity to Consume



Evaluated on $a \in [0, 1]$ using the spline-smooth Howard solution. **Left**: Consumption increases with a (concave) and with z (richer and more productive agents consume more). **Middle**: Net savings cross zero at the precautionary buffer stock. **Right**: MPC is highest near the borrowing constraint and declines sharply — a hallmark of hand-to-mouth behavior at low wealth.

Problem 2: Euler Equation Residuals



Unconstrained Euler condition:

$$c(a,z)^{-\alpha} = \beta(1+r)\,\mathbb{E}\big[c(a',z')^{-\alpha}\big]$$

Define consumption-equivalent error:

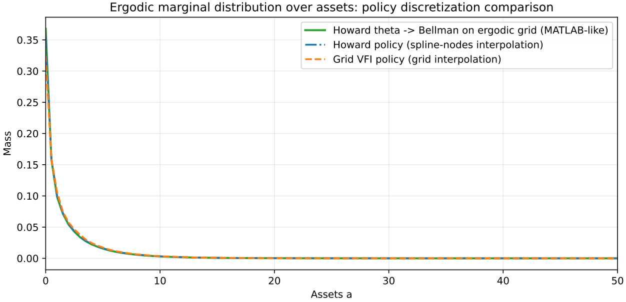
$$\hat{c}(a,z) = \big[\beta(1+r)\,\mathbb{E}c'^{-\alpha}\big]^{-1/\alpha}$$

Results (collocation nodes)
Max $|\hat{c}-c|/c$ on unconstrained: `2.3e-01`

Interpretation

Large Euler errors at the sparse solve-grid collapse on a denser output grid.

Problem 2: Ergodic Distribution



Three policy discretization methods:

1. **theta→Bellman**: recompute $a'(a, z)$ on ergodic grid from Howard θ (MATLAB-like)
2. **Spline nodes**: interpolate Howard policy via spline
3. **Grid VFI**: interpolate Grid-VFI policy

Aggregate moments

	Howard
Agg. assets	1.72–1.89
Agg. income	1.15
Agg. consump.	1.15
Stationarity err	`3.5e-17`

Note: Moments differ across

Method Comparison: Lessons from Both Problems

	Grid VFI	**Coeff VFI**	**Howard**
Convergence	Slow (300+)	Slow (same)	Fast (6–7)
Accuracy	Benchmark	$\approx$ Grid	$\approx$ Grid
Solve grid	Dense grid	Greville nodes	Same
Robustness	Robust	Basis-sensitive	Needs guess
Wall time	2.6 s	45 s	<1 s

Main lesson

Howard acceleration reduces Bellman maximizations to single digits while preserving full coefficient-based accuracy.

Diagnostics: What to Check After Solving

Solution quality

1. **Bellman residual**: $\|V^{t+1} - V^{t}\|$ at convergence
2. **Cross-check**: Grid VFI vs Howard on a dense evaluation grid
3. **Policy monotonicity**: $a'(a)$ non-decreasing in a
4. **Boundary**: $c = y + (1+r)a$ at the constraint

Distribution quality

1. **Stationarity**: $\|n - nQ\|$ (P1: `1.3e-17`; P2: `3.5e-17`)
2. **Row-sum of Q**: each row $= 1$
3. **Euler residual**: on dense output grid
4. **Mass at boundary**: $a_{\max}$ should not be binding

Rule of thumb

Convergence is *necessary* but not *sufficient*. Always verify residuals and moments.

Summary and Takeaways

Problem 1 (discrete income)

- Howard converges in 7 iterations vs. 336 (Grid VFI) and 420 (Coeff VFI) — same fixed point
- Unemployed agents hold less wealth; distribution mass concentrated at low a
- Stationarity check passes at machine precision

Problem 2 (continuous AR(1))

- Greville collocation nodes $\neq$ output grid; always re-evaluate on uniform grid before computing moments
- MPC decreasing in a: hand-to-mouth behavior at the borrowing constraint
- Euler errors on the sparse solve grid are misleading — use a dense output grid for residual reporting

Code: `examples/pset3/notebooks/pset3_problem1.ipynb` and `pset3_problem2.ipynb`

Appendix

Appendix: Calibration

Problem 1: Two-State Income

Symbol	Description	Value
β	Discount factor	0.95
r	Interest rate	0.04
b	Unemployment income	0.25
σ	Job-loss rate	0.10
λ	Job-finding rate	0.25
$a_{\max}$	Asset upper bound	10
n_{breaks}	Spline breakpoints	51
n_{nodes}	Greville nodes	53
K	Distribution grid	551

Problem 2: AR(1) Income

Symbol	Description	Value
β	Discount factor	0.95
r	Interest rate	0.02
α	CRRA coefficient	1.5
ρ	AR(1) persistence	0.90
σ	AR(1) innovation std. dev.	0.20
k	Quadrature nodes	7
$a_{\max}$	Asset upper bound	50
n_a	Asset breakpoints	51
n_z	Income breakpoints	15

Appendix: Howard Improvement — Linear System (Problem 1)

With fixed policy $a_e'(a)$ and $a_u'(a)$, the value functions satisfy:

$$\begin{pmatrix} \Phi - \beta(1-\sigma)\Phi(a_e') & -\beta\sigma\Phi(a_u') \\ -\beta\lambda\Phi(a_e') & \Phi - \beta(1-\lambda)\Phi(a_u') \end{pmatrix} \begin{pmatrix} \theta_e \\ \theta_u \end{pmatrix} = \begin{pmatrix} u(c_e) \\ u(c_u) \end{pmatrix}$$

where $\Phi = \Phi(a_{\text{nodes}})$ is the $K \times K$ basis matrix and $\Phi(a_e')$ evaluates the basis at policy $a_e'(a_{\text{nodes}})$.

For Problem 2 (single state variable z, 901 collocation nodes):

$$\left[\Phi - \beta \sum_{k=1}^{K} w_k \Phi(a'(\cdot),\, \rho z + \sigma\varepsilon_k)\right] \theta = u(c)$$

where $\Phi(a', z')$ is the 901×901 tensor-product basis matrix evaluated at next-period states.

Computational note

Both systems are **dense** and solved by LU factorization. Problem 1: 1102×1102 block; Problem 2: 901×901. This motivates using iterative policy updates rather than forming the full matrix at every step.